Brian M. Howell

computational optimization | *numerical methods* | *high performance computing*
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Summary

I am an engineer interested in high-performance numerical simulation/optimization/linear algebra and its applications in physics and AIML.

Professional Experience

Apple, Cupertino, CA. **Feb 2024 - present**
Optimization/ML Research Engineer

- ***Platform Architecture*** - I work on a team focusing on parallel/distributed for high-performance computing tools for internal use.
- ***AI***: I lead the infrastructure development of an internal AI engine for accelerating forward-modeling workflows.
- ***Optimization***: I also led the development of an internal optimization engine for large-scale distributed computing tasks related to high performance computing and machine learning.
- ***GPU kernel engineering***: Parallel computing via GPUs and MPI, low-level kernel development and performance optimization, large-scale distributed computing for numerical optimization, machine learning for low-rank accelerated modeling.

Google X, the moonshot factory, Mountain View, CA. **Jan 2022 - Dec 2022**
AI Resident:

- ***My project***aimed at bringing modern computing tools for materials optimization to a very large industry. My colleague and I cracked the problem with geometric/thermodynamic + convex modeling/optimization.
- ***Machine Learning/Optimization***:Gaussian processes + Bayesian optimization, deep learning, convex optimization
- ***Physics Simulation/Modeling***: Discrete element method, convex geometry
- ***Hardware***: Sensor development and data processing, high-throughput experimentation, feedback control systems for complex fluid flow
- ***Publicly Available Output***: Three patents granted (one as lead inventor)

Lawrence Livermore National Lab., Livermore, CA. **June 2017 - Jan 2022**
Staff Scientist:

- ***My work*** at LLNL was primarily focused on materials development & optimization for 3D printing
- ***Software/Simulation***: Controllers, sensors, toolpath generation and optimization, digital twins for additive manufacturing
- ***Hardware/Chemical***: Hardware integration, CAD modeling & design, chemical formulation
- ***Testing***: Rheology & UV kinetics, black (Instron), Scanning Electron Microscope (SEM)
- ***Publicly Available Output***: Two patents (one as lead inventor), one publication, work featured in Advanced Science News

Skills

Programming Tools: C/C++, Python, CUDA, METAL, OpenMP, MPI, PyTorch, JAX, MLX, Git, Linux
Computational Methods: Numerical Methods/Optimization/Linear Algebra, Machine Learning, Parallel Computing, Distributed Computing

Education

UC Berkeley
PhD/MS in **Computational/Mechanical Engineering** **2019-2024**
Dissertation: *Simulation Driven Optimization and Machine Learning for Advanced Manufacturing*
Advisor: *Prof. Tarek Zohdi*

Brigham Young University
BS in **Chemical Engineering** **2013-2017**